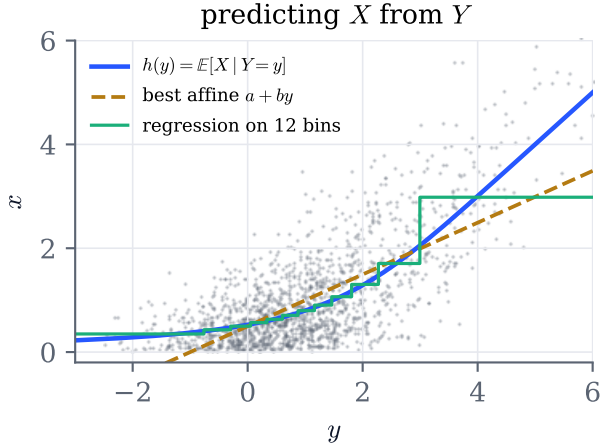
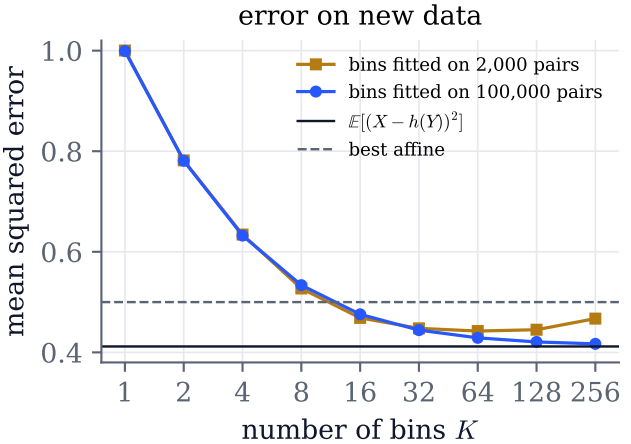

predicting X from Y
$h(y) = \mathbb{E}[X \mid Y = y]$
best affine $a + by$
regression on 12 bins
x
y
6
4
2
0
−2
0
2
4
6


error on new data
bins fitted on 2,000 pairs
bins fitted on 100,000 pairs
$\mathbb{E}[(X - h(Y))^2]$
best affine
mean squared error
number of bins K
1.0
0.8
0.6
0.4
1
2
4
8
16
32
64
128
256